universität innsbruck

Initial Presentation: Segfault Slayer

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Game Theme

Game Idea:

- Metroidvania-style 2D side-scroller
- Player fights programming concepts and bugs (mainly C++ related)
- Playful, Pixel-theme (e.g. ASCII-Art for textures)

World design:

- First Area: Hardware with Logic Operators
- Second Area: Programming with loops and other concepts

STUDENT: ALEX | HP: [#####] 5/6 | LEVEL 02
MAP [#####.##]
FOR LOOP
for(i=0;i<N;i++)
FUNCTION
int calculate(x,y)
( )
{ return x+y;
}

Game Sprites

Sprite Creation with Pixellab

We use **Pixellab**, a generative AI tool for creating game sprites.

It includes builders for:

- Characters
- Maps
- Animations

Project Structure

Directory Layout:

- `assets/`
- `build/`
- `data/`
- `docs/`
- `saves/`
- `src/`
  - `core/`
  - `entities/`
  - `rendering/`
  - `utils/`
  - `world/`
- `tests/`

State Loading/Unloading

Main dynamic JSON-File

- player-data (currency, xp, ...)
- room-data (collectables and enenmy booleans)

Loading

- loading static room json
- overwriting with state json

Unloading

- Fixed savepoints for resetting room-data

Development Setup

Dev.-Stack:

- Must: git, cmake, clang-format, vcpkg
- CI with clang-tidy
- 1 Windows & 2 Linux
- Assets: AI & manual polishing

Project Management:

- Issues
- Milestones
- Weekly Meetings

Dependencies

- SFML
- tilson
- spdlog / plog
- Catch2
- imgui (maybe)
- glaze (maybe)

Team Responsibilities

Task Distribution:

- (2) enemies - Lukas
- (1) menus - Lukas
- (2) two consecutive areas - Emanuel
- (1) save / load with dedicated save points (rooms) - Emanuel
- (1) map - Emanuel
- (2) basic movement - Dominik
- (1) one or more advanced movement mechanics - Dominik
- (1) main combat - Dominik